A COUNTEREXAMPLE TO GENERIC INJECTIVITY OF THE TANGENT GAUSS MAP

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Contents

1. Strategy 1
2. A dimension count for deformations 1
References 2

Theorem 0.1. *There exists a nondegenerate curve C on a simple abelian variety A of dimension $n > 2$ such that ϕ_C is not generically injective and no nontrivial automorphism $\tau \in \mathrm{Aut}(A)$ preserves C and acts as a scalar on T_0A.*

1. Strategy

Let C be a nonhyperelliptic curve, and let $\mathcal{L}$ be a line bundle of degree 3 with $h^0(\mathcal{L}) = 2$. Then $\mathcal{L}$ is base-point-free and induces a map $|\mathcal{L}| : C \longrightarrow \mathbb{P}^1$ of degree 3.

The linear system $\left(\mathcal{L}^{\otimes 2}, \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L})\right)$ induces a map $\phi : C \longrightarrow \mathbb{P}^2$ that factors through $|\mathcal{L}|$. Thus ϕ is not generically injective. Our aim is to realize ϕ as the tangent Gauss map of a curve $C \subset A$. This requires:

- $\mathcal{L}^{\otimes 2} = \omega_C$;
- $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) \subset \mathrm{H}^0(\omega_C)$ is the cotangent space of an abelian subvariety.

We refer to the second requirement as the *cotangent-space condition*.

If we first drop the nonhyperellipticity assumption, we can find a curve satisfying these requirements. Take a hyperelliptic curve C that admits a tangential cover $f : C \longrightarrow E$ at a Weierstrass point p_0. This means that the subspace $\mathrm{Sym}^2 \mathrm{H}^0(\omega_C(-2p_0)) \subset \mathrm{H}^0(\omega_C)$ satisfies the cotangent-space condition. For $g = 4$, set $\mathcal{L} = \mathcal{O}_C(3p_0)$. Then

- $\mathcal{L}^{\otimes 2} = \mathcal{O}_C(6p_0) \cong \omega_C$;
- $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}) \cong \mathrm{H}^0\left(\mathcal{O}_C(4p_0)\right) = \mathrm{H}^0\left(\omega_C(-2p_0)\right)$ satisfies the cotangent-space condition.

We use this hyperelliptic curve as a seed and deform it to obtain a nonhyperelliptic curve while preserving the two requirements above.

2. A dimension count for deformations

We seek a local deformation $(C_t, \mathcal{L}_t)$ of the initial spin curve for which $\mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) \subset \mathrm{H}^0(\omega_{C_t})$ continues to satisfy the cotangent-space condition and C_t is nonhyperelliptic.

Let B denote the local deformation space of the spin curve $(C_0, \mathcal{L}_0)$. Its dimension is

$$\dim B = h^1(\mathcal{T}_C) = 3g - 3 = 9.$$

(1) The locus $\Theta_{null} = \left\{(C, \mathcal{L}) \;\middle|\; h^0(\mathcal{L}) \geqslant 2\right\}$ is a divisor.

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(2) Define the rational subspace

$$V_0 := H^1(C_0;\mathbb{Q}) \cap \mathrm{H}^0\left(\omega_C(-2p_0)\right) \subset H^1(C_0;\mathbb{Q}),$$

and let $V_t \subset H^1(C_t;\mathbb{Q})$ be its parallel transport under the Gauss–Manin connection.

We also define the complex subspace

$$W_0 := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_\prime) \subset \mathrm{H}^0(\omega_{C_0}),$$

and let $W_t := \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_\sqcup) \subset \mathrm{H}^0(\omega_{C_t})$ be the corresponding subspace obtained by deforming the line bundle.

To ensure that W_t continues to satisfy the cotangent-space condition, we use the following equivalent formulations:

$$\begin{aligned}
&\begin{cases} V_t \text{ remains a Hodge structure} \\ V_{t,\mathbb{C}} \cap \mathrm{H}^0(\omega_C) = W_t \end{cases} \\
\iff\ & V_{t,\mathbb{C}} = W_t \oplus \overline{W}_t \\
\iff\ & W_t \subseteq V_{t,\mathbb{C}} \\
\iff\ & \mathrm{rk}\left[W_t \longrightarrow H^1(C_t;\mathbb{C})/V_{t,\mathbb{C}}\right] = 0
\end{aligned}$$

The last condition cuts out a locus of codimension 6. Together with the divisor condition above, this leaves a deformation space of dimension $9 - 1 - 6 = 2$.

(3) To find a nonhyperelliptic deformation, we compare this dimension with that of the local hyperelliptic branch. On a hyperelliptic curve C_t, a line bundle with $\deg \mathcal{L}_t = 3$ and $h^0(\mathcal{L}_t) = 2$ must satisfy $\mathcal{L}_t \cong \mathcal{O}_{C_t}(3p_0)$ for some Weierstrass point p_0. Consequently,

$$W_t = \mathrm{Sym}^2 \mathrm{H}^0(\mathcal{L}_t) = \mathrm{H}^0\left(\omega_{C_t}(-2p_0)\right).$$

The following two lemmas show that the local hyperelliptic branch has dimension 1:

Lemma 2.1. *Let C be a hyperelliptic curve and p_0 a Weierstrass point. The subspace $\mathrm{H}^0\left(\omega_{C_t}(-2p_0)\right)$ satisfies the cotangent-space condition if and only if there exists a tangential cover $f : C \longrightarrow E$ at p_0.*

Lemma 2.2. [1, Proposition 2.5, 3] *For a fixed elliptic curve E and an integer d, there are only finitely many tangential covers $f : C \longrightarrow E$ of degree d.*

Thus, within a 3-dimensional deformation space, the hyperelliptic locus has dimension at most 2. It follows that there exists a nonhyperelliptic curve satisfying the required conditions.

References

[1] Armando Treibich. Hyperelliptic tangential covers and even elliptic finite-gap potentials, back and forth, 2025.

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